Hybrid Image Encryption Using RSA, OAEP and AES: Implementation and Analysis

Yasmina Bellouz - Aya Mokhnache

Abstract

This work presents a comparative study of three image encryption approaches: RSA, RSA with OAEP padding, and a hybrid RSA-OAEP-AES scheme. The main objective is to analyze their security strength and computational efficiency when applied to digital images. The methods are evaluated using histogram analysis, correlation coefficients,entropy, and execution time measurements. Experimental results show that RSA-OAEP improves randomness compared to basic RSA, while the hybrid RSA-OAEP-AES approach achieves the best performance in terms of speed, scalability, and cryptographic security, making it more suitable for practical image encryption applications.

1 Introduction

Nowadays, digital image exchange has become very common in many domains such as medical systems, military communications, cloud storage, and multimedia applications. However, image data transmitted through unsecured channels may be intercepted, modified, or accessed by unauthorized users. Therefore, cryptography plays an important role in protecting sensitive image information.

Cryptographic systems are divided into symmetric and asymmetric cryptosystems. Symmetric algorithms such as AES use the same secret key for encryption and decryption and are generally very fast. Asymmetric algorithms such as RSA use two different keys: a public key for encryption and a private key for decryption.

RSA is one of the most widely used public-key cryptosystems. Its security is based on the difficulty of factoring large integers. However, RSA suffers from high computational complexity when encrypting large image data directly. Furthermore, plain RSA encryption is deterministic and therefore vulnerable to several cryptographic attacks.

To improve RSA security, padding schemes such as OAEP (Optimal Asymmetric Encryption Padding) are introduced to provide semantic security. In addition, hybrid encryption systems combining RSA and AES are commonly used in practice to benefit from both the strong security of RSA and the high speed of AES.

In this work, three image encryption approaches are implemented and compared:

1. RSA image encryption
2. RSA with OAEP padding
3. Hybrid RSA-OAEP with AES

2 Implementation of the three tasks

The implementation was developed using Python and several image-processing and cryptographic libraries.Before going through details, we clarify the following : We distinguish three types (modes) of images:

- **Grayscale images:** Pixels represent intensity levels ranging from black to white.
- **RGB:** Each pixel contains three channels: Red, Green, and Blue.
- **RGBA:** Same as RGB but with an additional alpha channel representing transparency.

- We treat each mode separately . As banchmarks images we use the following:



Figure 1: Original images

RF

2.1 Task 01: RSA Image Encryption

The first implementation applies RSA directly to image encryption and decryption. The image pixels are flattened into a one-dimensional sequence and each pixel value is converted into its 8-bit binary representation. RSA public and private keys are then generated by selecting two prime numbers and computing the RSA parameters (n, e, d). As primality test we use Miller Rabin to ensure that p and q are primes

The block size is determined according to the RSA modulus size using:

$$\text{Block Size} = \left\lfloor \frac{\log_2(n)}{8} \right\rfloor$$

The binary pixel sequence is divided into blocks whose sizes are compatible with the RSA modulus. Each binary block is converted into a decimal integer before encryption.

The RSA encryption function is then applied to each image block individually using the public key (e, n). The encrypted blocks are finally reconstructed into a visual encrypted image in order to observe the effect of encryption on the image structure.

For decryption, each encrypted block is decrypted using the RSA private key,for optimality we apply the CRT The recovered numerical blocks are converted back into binary form and divided again into individual 8-bit pixel values. Finally, the recovered pixels are reshaped according to the original image dimensions in order to reconstruct the decrypted image.

2.1.1 Results and Interpretation:

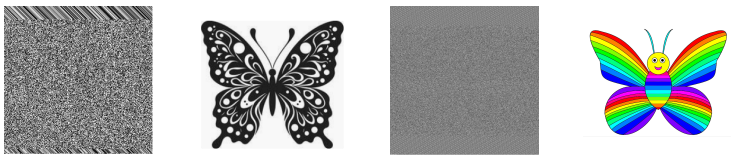


Figure 2: Task 01 results

The encrypted images shown in Figure 2 exhibit highly random and noise-like appearances, making the original visual information completely unintelligible. The significant modification of pixel values demonstrates the effectiveness of the RSA encryption algorithm in concealing image content and resisting direct visual analysis attacks.

For both the black-and-white and colored images, the encrypted results show no recognizable structures or patterns from the original images. The colored encrypted image especially illustrates the strong alteration of RGB channel information during the encryption process.

The decrypted images are visually identical to the original images, confirming the correctness of the RSA decryption procedure and the successful reconstruction of the original pixel data without significant information loss.

Some minor visual artifacts may appear in the encrypted images due to the direct block-based RSA processing and the reconstruction operations applied during image encryption and decryption.

2.2 Task 02: RSA with OAEP Padding

The second implementation improves the security of direct RSA image encryption by integrating OAEP (Optimal Asymmetric Encryption Padding) before the encryption process. Unlike direct RSA encryption, the block size must now take into account the additional OAEP padding information. The block size is computed according to the RSA modulus size and the OAEP overhead:

$$\text{Block Size} = k - 2hLen - 2$$

where:

- k is the RSA modulus size in bytes,
- $hLen$ is the output length of the SHA-256 hash function.

The binary image sequence is divided into RSA-compatible blocks and converted into decimal integers. Since OAEP operates on byte sequences, each numerical block is converted into bytes before padding.

The OAEP encoding process consists of:

1. Creating a data block containing the hashed label, padding string, separator byte, and message
2. Generating a random seed using secure random bytes
3. Applying mask generation functions (MGF) based on SHA-256
4. Masking both the data block and the random seed using XOR operations
5. Constructing the final OAEP encoded message

After OAEP encoding, the padded blocks are converted into integers and encrypted using RSA public-key encryption.

For decryption, RSA decryption is first applied to recover the OAEP-encoded blocks. The recovered blocks are converted back into byte sequences and passed through the OAEP decoding procedure. The decoding process removes the padding, verifies the SHA-256 hash consistency, and extracts the original message blocks.

The recovered blocks are converted back into binary pixel sequences and reshaped according to the original image dimensions in order to reconstruct the decrypted image.

Compared with Task 01, this implementation provides stronger semantic security and improved randomness properties due to the randomization introduced by OAEP padding.

2.2.1 Results and Interpretation:

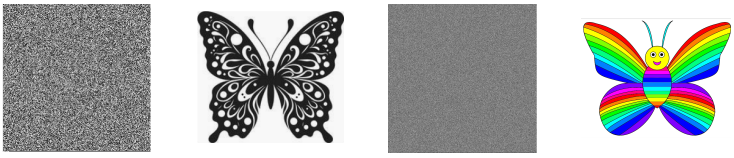


Figure 3: Task 02 results

The encrypted images produced using RSA-OAEP exhibit highly random and noise-like appearances with no visible traces of the original image structures. Compared with the direct RSA encryption results of Task 01, the encrypted images generated by RSA-OAEP show better randomness and more uniform pixel distributions for both the black-and-white and colored images.

This improvement is mainly due to the OAEP padding scheme, which introduces randomization before RSA encryption and therefore enhances semantic security. As a result, identical image blocks no longer produce identical ciphertext blocks, improving resistance against visual cryptanalysis attacks.

The encrypted colored image demonstrates a strong alteration of the RGB channel information, while the grayscale encrypted image shows a highly uniform random texture with no recognizable patterns.

The decrypted images are visually identical to the original images, confirming the correctness of the RSA-OAEP decryption process and the successful removal of the OAEP padding after RSA decryption without significant information loss.

2.3 Task 03: Hybrid RSA-OAEP with AES

The third implementation combines RSA-OAEP and AES in order to build a hybrid cryptographic system that provides both high security and fast execution speed. Unlike the previous approaches, RSA is no longer used to encrypt the entire image directly. Instead, AES is used for bulk image encryption while RSA-OAEP securely encrypts the AES secret key.

AES (Advanced Encryption Standard) is a symmetric encryption algorithm optimized for fast encryption of large amounts of data. AES operates on fixed-size blocks and supports different key sizes such as 128, 192, and 256 bits. In this implementation, a 256-bit AES secret key is generated randomly using secure random-byte generation:

$$\text{AES Key Size} = 256 \text{ bits}$$

AES is used in CBC (Cipher Block Chaining) mode, which improves security by ensuring that identical plaintext blocks do not produce identical ciphertext blocks. CBC mode requires an Initialization Vector (IV), which is generated automatically during encryption.

Since AES operates on fixed-size blocks, padding is applied to the image byte sequence in order to ensure that the data length is compatible with the AES block size.

The AES encryption process consists of:

1. Generating a random AES secret key
2. Generating a random Initialization Vector (IV)
3. Padding the image byte sequence
4. Encrypting the image using AES-CBC mode

The encrypted AES ciphertext is then reconstructed into a visual encrypted image in order to analyze the effect of encryption on the image structure.

After encrypting the image with AES, RSA-OAEP is used for secure key exchange. The AES secret key is converted into an integer

and encrypted using RSA public-key encryption. At the receiver side, the AES key is recovered using RSA private-key decryption.

Once the AES secret key is recovered, AES decryption is applied using the same CBC mode and Initialization Vector. The decrypted byte sequence is then unpadded and reshaped according to the original image dimensions in order to reconstruct the decrypted image.

Compared with direct RSA encryption, this hybrid implementation significantly reduces computational cost because AES efficiently encrypts large image data while RSA-OAEP protects only the AES secret key. Consequently, the hybrid RSA-OAEP-AES approach achieves better scalability, higher execution speed, and strong cryptographic security.

2.3.1 Results and Interpretation:

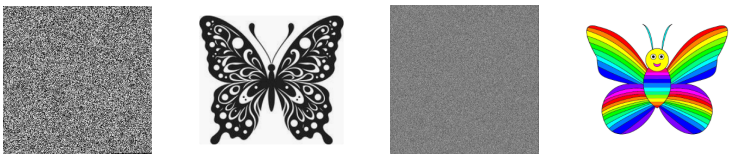


Figure 4: Task 02 results

The encrypted images generated using the hybrid RSA-OAEP-AES approach exhibit highly random and noise-like appearances with no visible traces of the original image information. For both the grayscale and colored images, the encryption process successfully conceals the visual structures and significantly modifies the pixel distributions.

The encrypted colored image demonstrates strong alteration of the RGB channel information, while the grayscale encrypted image presents a highly uniform random texture with no recognizable patterns. This behavior indicates strong resistance against attacks.

The decrypted images are visually identical to the original images, confirming the correctness of both the RSA-OAEP key recovery process and the AES decryption operation without significant information loss.

Compared with the previous approaches, the hybrid RSA-OAEP-AES solution provides the best balance between randomness, execution speed, computational efficiency, and cryptographic security, making it the most suitable approach for practical large-scale image encryption applications.

3 COMPLEXITY AND PERFORMANCE ANALYSIS

Table 1: Complexity and Performance Comparison

Method	Complexity	Relative Speed
RSA	$\mathcal{O}(n^3)$	Slow
RSA + OAEP	$\mathcal{O}(n^3) + \mathcal{O}(M)$	Moderate
RSA + OAEP + AES	$\mathcal{O}(n^3) + \mathcal{O}(Mr)$	Fast

Where : n is the RSA key size in bits, M is the total image data size in bytes, r is the number of AES rounds (fixed at 10 for AES-256)

Table 2: Execution Time Comparison

Method	Colored image	Black and white image
RSA	243.27814388275146	17.13881540298462
RSA + OAEP	316.9228973388672	30.314766883850098
RSA + OAEP + AES	8.742405891418457	7.956896066665649

3.0.1 Interpretation:

The execution-time results are consistent with the theoretical complexity analysis presented in Table I. Direct RSA image encryption has a computational complexity of approximately $\mathcal{O}(n^3)$ due to the modular exponentiation operations performed on image blocks. Consequently, RSA requires very large execution times, especially for colored images containing large amounts of pixel data.

The RSA-OAEP approach introduces additional padding and randomization operations with complexity $\mathcal{O}(M)$. As a result, the total complexity becomes approximately $\mathcal{O}(n^3) + \mathcal{O}(M)$, leading to slightly higher execution times compared with direct RSA encryption. However, this additional cost significantly improves semantic security and randomness properties.

The hybrid RSA-OAEP-AES solution achieves the best performance despite combining multiple cryptographic algorithms. Its complexity is approximately $\mathcal{O}(n^3) + \mathcal{O}(Mr)$. In this approach, RSA-OAEP is applied only to the AES secret key, while AES performs the bulk image encryption. Since AES is highly optimized for large-volume data encryption, the execution times are adically reduced compared with the previous approaches.

The results also show that colored images require more execution time than black-and-white images because RGB images contain multiple color channels and a larger quantity of pixel data. Overall, the hybrid RSA-OAEP-AES approach provides the best compromise between computational complexity, execution speed, and cryptographic security, making it the most suitable solution for practical image encryption applications.

4 Security Strength Evaluation

4.1 Histograms

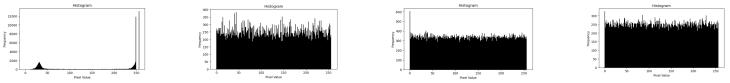


Figure 5: Histograms for uncolored image

- **Original image:** The histogram is highly concentrated around specific intensity (white and black), showing strong redundancy and predictable grayscale structure typical of natural images.
- **RSA:** The histogram becomes more distributed across the intensity range, indicating partial randomization of pixel values and reduced statistical predictability.
- **RSA + OAEP:** The distribution appears more uniform than RSA alone due to OAEP random padding, improving resistance against histogram-based statistical attacks.
- **RSA + OAEP + AES:** The histogram is almost perfectly flat and noise-like, confirming maximum diffusion and highly secure pixel randomization.

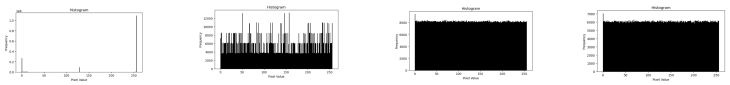


Figure 6: Histograms for colored image

- **Original image:** Multiple visible peaks appear across RGB intensity levels, reflecting structured color information and high channel dependency.
- **RSA:** The histogram becomes more irregular and randomized, although some residual peaks remain due to incomplete diffusion in color channels.

- **RSA + OAEP:** OAEP significantly smooths the histogram and removes most remaining statistical structures through randomized padding.

- **RSA + OAEP + AES:** The histogram becomes nearly perfectly uniform, indicating complete destruction of color patterns and excellent encryption quality.

4.2 Scatter Plots

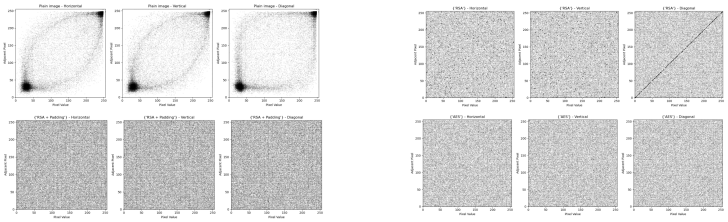


Figure 7: Scatter plots for uncolored image

- **Original image:** A strong diagonal structure is visible, confirming very high correlation between adjacent grayscale pixels and strong spatial redundancy.

- **RSA:** The diagonal pattern becomes weaker and more dispersed, indicating partial decorrelation and increased randomness after encryption.

- **RSA + OAEP:** The scatter points become highly randomized with almost no visible structure, confirming near-zero pixel dependency.

- **RSA + OAEP + AES:** The distribution becomes fully uniform across the scatter space, demonstrating complete elimination of spatial correlation.

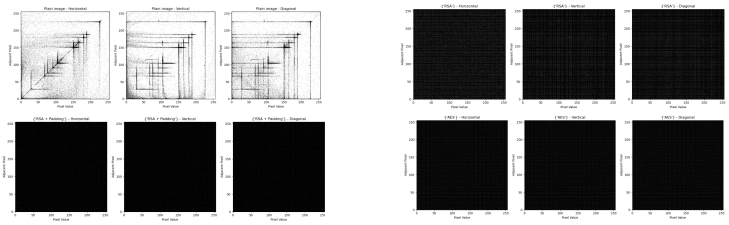


Figure 8: Scatter plots for colored image

- **Original image:** Clear linear clustering appears across RGB channels, reflecting strong inter-pixel dependency in natural color images.

- **RSA:** The scatter plot becomes significantly more random, showing that RSA effectively reduces adjacent-pixel correlation.

- **RSA + OAEP:** The point distribution becomes almost entirely structureless, confirming stronger decorrelation due to OAEP randomization.

- **RSA + OAEP + AES:** A perfectly uniform random cloud is obtained, indicating complete decorrelation across all color channels.

4.3 Correlation Coefficients

The correlation analysis produced similar observations for both grayscale and colored images.

Table 3: Grayscale Image

Direction	Plain	RSA	RSA+Padding	AES
Horizontal	0.939855	0.012952	0.008361	0.011774
Vertical	0.943197	0.004876	0.000236	-0.002963
Diagonal	0.897751	0.117849	0.001917	0.002506

Table 4: Color Image

Direction	Plain	RSA	RSA+Padding	AES
Horizontal	0.957811	0.035493	0.001190	0.001173
Vertical	0.929806	0.005953	0.001173	0.000306
Diagonal	0.916154	-0.038807	0.000550	0.000498

- **Original Images:** Correlation coefficients remain very close to +1 in all directions for both grayscale and colored images, confirming strong spatial redundancy and high dependency between neighboring pixels.

- **RSA:** Correlation values decrease significantly after encryption, indicating considerable reduction of pixel dependency, although some residual correlation may still remain.

- **RSA + OAEP:** All correlation coefficients become extremely close to zero, demonstrating strong decorrelation, improved randomness, and effective removal of remaining pixel dependency.

- **RSA + OAEP + AES:** Similar near-zero correlation values are obtained, confirming excellent decorrelation, high randomness, and strong resistance against statistical attacks. In addition, the hybrid approach provides better computational efficiency and stronger practical security.

4.4 Entropy Analysis

Table 5: Entropy Comparison

Image	Original	RSA	RSA+OAEP	RSA+OAEP+AES
Gray Image	5.4456	7.9862	7.9964	7.9971
Color Image	1.8202	7.9049	7.9999	7.9998

Entropy measures the randomness of encrypted pixels, where the ideal value for an 8-bit image is 8 bits/pixel. The original images exhibit relatively low entropy due to their natural redundancy and structured content.

All encryption methods significantly increase entropy above 7.9, indicating strong randomization and effective concealment of image information. The hybrid RSA-OAEP-AES approach achieves entropy values closest to the theoretical maximum, confirming superior diffusion capability and stronger resistance against entropy-based statistical attacks.

Table 6: Comparative Security Evaluation of Encryption Methods

Metric	RSA	RSA+OAEP	RSA+OAEP+AES
Histogram	Partially random	Highly random	Almost perfectly uniform
Scatter Plot	Partial decorrelation	Strong decorrelation	Complete decorrelation
Correlation Coefficient	Reduced correlation	Near-zero correlation	Ideal near-zero correlation
Entropy	High randomness	Very high randomness	Near-ideal randomness
Overall Security	Good	Very Good	Excellent

5 Conclusion

This work compared three image encryption approaches: RSA, RSA-OAEP, and hybrid RSA-OAEP-AES. Results show a clear progression in both security and efficiency. Plain RSA introduces randomness but remains deterministic and computationally slow. Adding OAEP eliminates residual statistical structure through probabilistic padding. The hybrid scheme achieves the best performance overall, reducing execution time compared to RSA-OAEP while delivering perfectly uniform histograms, complete spatial decorrelation, and near-zero correlation coefficients across all tested images. These results confirm that combining asymmetric key exchange with symmetric bulk encryption is the most practical and cryptographically sound solution for real-world image encryption.